

Response to the Reviewers

We thank both reviewers for their careful assessment of the manuscript. We are grateful that Reviewer 1 confirmed the contribution, technical soundness, comprehensive presentation, and suitability of the references. We also thank Reviewer 2 for the constructive comments, which helped us sharpen the scope of the contribution and avoid overclaiming. The revised manuscript now frames the work explicitly as a controlled simulation benchmark of deep reinforcement learning with runtime safety monitoring and not as proof of operationally safe on-orbit autonomy.

Note on the highlighted manuscript. In the highlighted version of the manuscript, the changes made in response to Reviewer 2 are marked in yellow. Each marked passage begins with a tag in square brackets, for example [R2.1]. The tag refers to the numbered concern of Reviewer 2 in this letter (R2.1 to R2.5). Where only single words of a sentence were changed, the tag stands at the beginning of the sentence and only the changed words are highlighted. Revisions that do not relate to a specific concern of the reviewers are not highlighted. These are mainly language editing, corrections of individual values and references, and updated model diagrams. The running head was changed from “Safe RL for Free-Floating Space Robot Manipulators” to “Benchmarking Deep RL With Runtime Safety Monitoring for Free-Floating Space Manipulators” (R2.5). This change cannot be highlighted in the page header.

Response to Reviewer 1

We thank the reviewer for the positive evaluation of the manuscript and for confirming that the paper contributes to the body of knowledge, is technically sound, presents the subject matter comprehensively, and uses applicable and sufficient references. No specific changes were requested by this reviewer.

Response to Reviewer 2

We thank the reviewer for the detailed and constructive comments. We agree that the contribution should be understood as a benchmarking and integration study and not as a new reinforcement learning algorithm, control-theoretic method, or a formal proof of safe on-orbit control. We have revised the manuscript accordingly and have bounded the claims throughout the title, abstract, safety section, discussion, limitations, and conclusion.

Narrowing of the safety and formal-verification claims (R2.1). We revised the safety-related framing to avoid any implication of full formal safety certification. The main changes were made in the Abstract and in Section V, especially the subsection “Scope of the Safety Components.” The revised manuscript now states that the safety component consists of runtime monitors for selected properties and a formal check of torque saturation, and not a formal proof of closed-loop safety. The safety section further clarifies that the mechanisms cover collision events, joint-limit events, bounded torques, and momentum consistency, but that only the torque bound is checked formally. The collision and joint-limit monitors detect violations at runtime and terminate affected episodes. They do not prove absence of violations. The manuscript now also states explicitly that closed-loop stability, task completion, and collision avoidance over all possible trajectories of the full nonlinear Simscape model are not formally verified.

Clarification of the controlled simulation scope (R2.2). We revised Section III, “System Modeling and Simulation Environment,” to make the simulation assumptions explicit. The manuscript now states that the benchmark assumes ideal sensing, zero communication latency, rigid multibody dynamics, and no external disturbances in the main environment. It also states that the results characterize relative algorithm performance in a controlled simulation benchmark and do not constitute an assessment of operational readiness for on-orbit deployment. This limitation is repeated in the Abstract and in Section VIII, “Discussion and Conclusion.”

Expansion and bounding of the evaluation protocol (R2.3). We revised Section IV, “Reinforcement Learning Framework,” especially the evaluation protocol, to clarify the training and evaluation design. Each configuration is now trained with ten seeds, all ten seeds are reported, and no checkpoint is selected. The evaluation uses the same 31 initial states across agents, and the manuscript states that the conclusions remain conditional on these initial

states and the tested reference trajectories. In Section VIII, the limitations now explicitly state that the evaluation is limited to ten seeds, 31 initial states, one robot model, two reference paths, planar motion, and a small set of perturbations. The manuscript therefore no longer presents the results as a definitive ranking of all considered RL algorithms.

Reward ablation and reward-weight sensitivity (R2.4). We added analyses in Section VII, “Optimization of the PPO Agent,” to address the concern that the hand-tuned reward weights could bias the benchmark. The revised manuscript includes a reward-component ablation study, in which the optimized PPO agent is retrained with five seeds and one reward term disabled. It also includes a one-at-a-time sensitivity analysis of four reward weights, where each selected weight is multiplied by 0.5 or 2 and the agent is retrained with five seeds. These analyses make the influence of the reward design more transparent and bound the interpretation of the benchmark.

Robustness tests under non-ideal conditions (R2.2, R2.3). We extended Section VII-D, “Robustness Evaluation under Non-Ideal Conditions.” The trained policies are evaluated without retraining under observation noise, reduced torque limits, an external additive joint torque, uniformly increased masses, inertias, and joint damping coefficients, and a combined perturbation case. These tests do not remove the simulation-only limitation. In our opinion, they provide additional evidence beyond the nominal idealized benchmark.

Avoidance of broad safe-RL wording (R2.5). We revised the wording throughout the manuscript to avoid broad claims such as “safe RL” for the present contribution. The manuscript now consistently uses the more precise framing “runtime safety monitoring” and ties safety statements to the monitored properties and the torque-bound check. The conclusion states that the simulation results do not establish formal safety or operational readiness but support further investigation of deep RL with runtime safety monitoring for space manipulation tasks.

Summary of sections and subsections changed

Reviewer concern	Section/subsection highlighted in the revision	Change made
Formal-verification claim too broad (R2.1)	Abstract; Section V, “Scope of the Safety Components”	Safety claim narrowed to runtime monitors for selected properties and a formal torque-bound check; no formal proof of closed-loop safety is claimed.
Simulation assumptions and operational readiness (R2.2)	Section III, “System Modeling and Simulation Environment”; Section VIII, “Discussion and Conclusion”	Ideal sensing, zero latency, rigid-body dynamics, and absence of external disturbances are stated explicitly; operational readiness is excluded.
Evaluation size and conditional conclusions (R2.3)	Section IV, “Reinforcement Learning Framework”; Section VIII, “Limitations and Future Work”	Evaluation expanded and clarified with ten seeds, 31 initial states, no checkpoint selection, and conditional interpretation of rankings.
Reward-weight bias (R2.4)	Section VII, “Optimization of the PPO Agent”	Reward-component ablation and one-at-a-time reward-weight sensitivity analyses were added.
Robustness beyond nominal assumptions (R2.2, R2.3)	Section VII-D, “Robustness Evaluation under Non-Ideal Conditions”	Stress tests under observation noise, reduced torque limits, external joint torque, parameter scaling, and a combined case were added.
Broad “safe RL” wording (R2.5)	Abstract; Section V; Section VIII	Wording changed to “runtime safety monitoring” and safety statements tied to monitored properties and the torque-bound check.

Closing statement. We believe that these revisions directly address the reviewer's concerns while preserving the main contribution of the paper: a controlled benchmark of six deep reinforcement learning algorithms for Cartesian path tracking of a free-floating space robot manipulator, evaluated with unified metrics, runtime safety monitoring, PPO tuning, reward ablation and sensitivity analyses, and stress tests under selected non-ideal conditions.